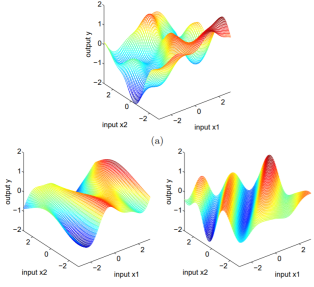
Introduction to Machine Learning

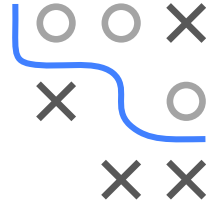
Gaussian Processes
Covariance functions for GPs



Learning goals

- Covariance functions encode key assumptions about the GP
- Know common covariance functions like squared exponential and Matérn

VALID COVARIANCE FUNCTIONS



- Recall marginalization property of GPs: for any $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\} \subset \mathcal{X}$,

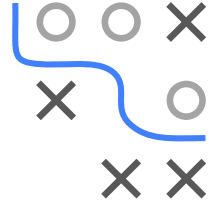
$$\mathbf{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- For $\mathbf{K}$ to be a valid cov matrix it needs to be positive semi-definite (PSD) for any choice of inputs $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$

- Cov function (or **kernel**) determines cov matrix: $\mathbf{K}_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$

- Implication: valid cov functions $k(\cdot, \cdot)$ must be PSD too

STATIONARY COVARIANCE FUNCTIONS



- Recall concept of spatial correlation

$$\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \text{ close in } \mathcal{X} \Rightarrow f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)}) \text{ close in } \mathcal{Y}$$

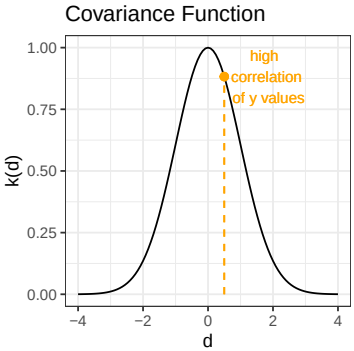
- Measure "closeness" via $\boldsymbol{d} = \mathbf{x}^{(i)} - \mathbf{x}^{(j)}$

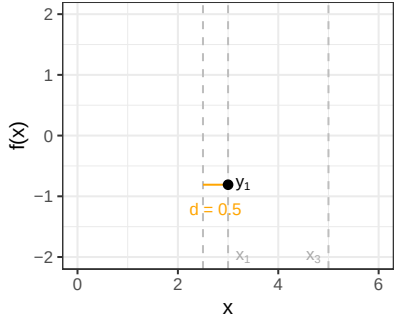
- $k(\cdot, \cdot)$ called **stationary** $\Leftrightarrow$ function of $\boldsymbol{d}$

$$k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = k(\boldsymbol{d})$$

EXAMPLE: STATIONARY COVARIANCE

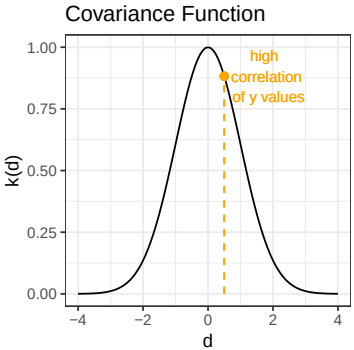
- Let $f(\mathbf{x}) \sim \mathcal{GP}(m, k)$ with $k(\mathbf{x}, \mathbf{x}') = \exp(-\frac{1}{2}\|\boldsymbol{d}\|^2)$
- Consider two points $\mathbf{x}^{(1)} = 3$ and $\mathbf{x}^{(2)} = 2.5$
- If you want to know about the correlation between $f(\mathbf{x}^{(1)})$ and $f(\mathbf{x}^{(2)})$, compute $\boldsymbol{d}(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$

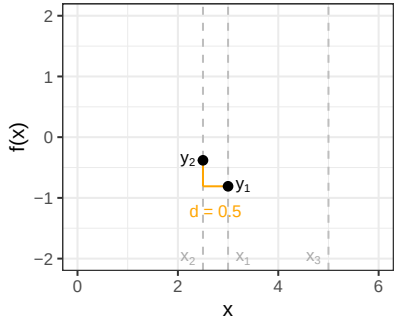




EXAMPLE: STATIONARY COVARIANCE

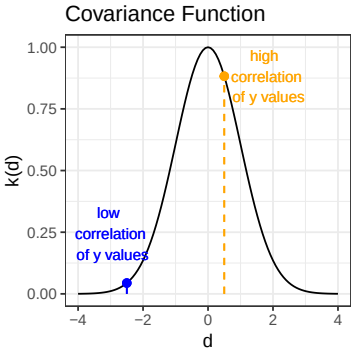
- Suppose we observe $y^{(1)} = -0.8$
- Under the above GP assumption, $y^{(2)}$ should be close to $y^{(1)}$

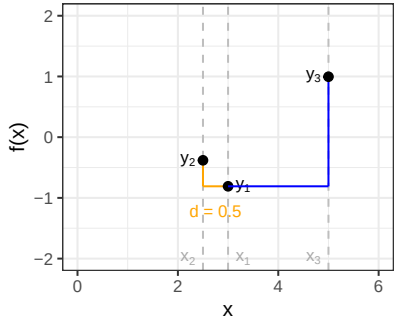




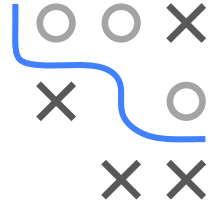
EXAMPLE: STATIONARY COVARIANCE

- Consider now $\mathbf{x}^{(3)} = 5$
- Further from $\mathbf{x}^{(1)} \Rightarrow$ expect lower correlation between $y^{(3)}$, $y^{(1)}$



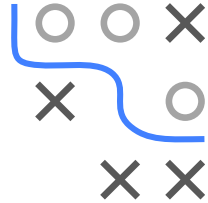


PROPERTIES OF COVARIANCE FUNCTIONS



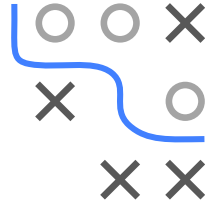
- Most cov functions belong to 3 common types
- **Stationary**, as discussed: $k = k(\boldsymbol{d})$
  $\Rightarrow$ invariant to translations in $\mathcal{X}$: $k(\mathbf{x}, \mathbf{x} + \boldsymbol{d}) = k(\mathbf{0}, \boldsymbol{d})$
- **Isotropic**: $k = k(r)$ with $r = \|\mathbf{x} - \mathbf{x}'\|$
  $\Rightarrow$ invariant to rotations of $\mathcal{X}$
- Isotropy implies stationarity
- **Dot product**: $k = k(\mathbf{x}^T\mathbf{x}')$

CONSTANT KERNEL



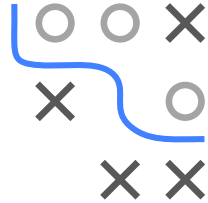
- intuition, formula, viz

LINEAR KERNEL



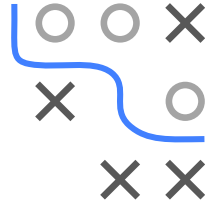
- intuition, formula, viz

POLYNOMIAL KERNEL



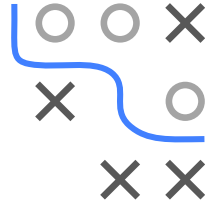
- intuition, formula, viz

MATÉRN KERNEL



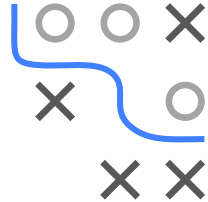
- intuition, formula, viz

PERIODIC KERNEL



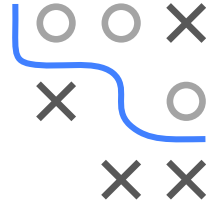
- intuition, formula, viz

EXPONENTIAL KERNEL



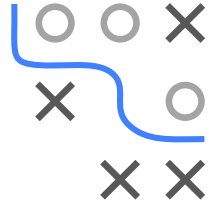
- intuition, formula, viz
- GPs with this kernel are called Ornstein-Uhlenbeck processes

SQUARED EXPONENTIAL KERNEL



- intuition, formula, viz

CHARACTERISTIC LENGTH-SCALE



- ...
- mention: componentwise generalization of SVM bandwidth hyperparam?

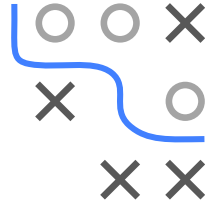
COMMONLY USED COVARIANCE FUNCTIONS

raus - cheatsheet

Name	$k(\boldsymbol{x}, \boldsymbol{x}')$
constant	σ_0^2
linear	$\sigma_0^2 + \boldsymbol{x}^T \boldsymbol{x}'$
polynomial	$(\sigma_0^2 + \boldsymbol{x}^T \boldsymbol{x}')^p$
squared exponential	$\exp(-\frac{\lVert \boldsymbol{x}-\boldsymbol{x}' \rVert^2}{2\ell^2})$
Matérn	$\frac{1}{2^\nu \Gamma(\nu)} \left(\frac{\sqrt{2\nu}}{\ell} \lVert \boldsymbol{x} - \boldsymbol{x}' \rVert \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{\ell} \lVert \boldsymbol{x} - \boldsymbol{x}' \rVert \right)$
exponential	$\exp\left(-\frac{\lVert \boldsymbol{x}-\boldsymbol{x}' \rVert}{\ell}\right)$

$K_\nu(\cdot)$ is the modified Bessel function of the second kind.

SQUARED EXPONENTIAL COVARIANCE FUNCTION



The squared exponential function is one of the most commonly used covariance functions.

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right).$$

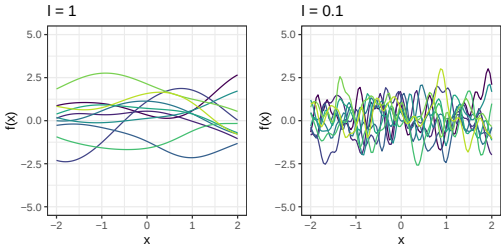
Properties:

- It depends merely on the distance $r = \|\mathbf{x} - \mathbf{x}'\| \rightarrow$ isotropic and stationary.
- Infinitely differentiable $\rightarrow$ sometimes deemed unrealistic for modeling most of the physical processes.

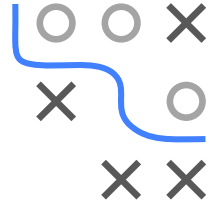
CHARACTERISTIC LENGTH-SCALE I

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2\ell^2}\|\mathbf{x} - \mathbf{x}'\|^2\right)$$

ℓ is called **characteristic length-scale**. Loosely speaking, the characteristic length-scale describes how far you need to move in input space for the function values to become uncorrelated. Higher ℓ induces smoother functions, lower ℓ induces more wiggly functions.



CHARACTERISTIC LENGTH-SCALE II



For $p \geq 2$ dimensions, the squared exponential can be parameterized:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2}\left(\mathbf{x} - \mathbf{x}'\right)^\top \boldsymbol{M}\left(\mathbf{x} - \mathbf{x}'\right)\right)$$

Possible choices for the matrix $\boldsymbol{M}$ include

$$\boldsymbol{M}_1 = \ell^{-2}\boldsymbol{I} \qquad \boldsymbol{M}_2 = \mathrm{diag}(\boldsymbol{\ell})^{-2} \qquad \boldsymbol{M}_3 = \Gamma\Gamma^\top + \mathrm{diag}(\boldsymbol{\ell})^{-2}$$

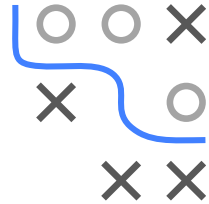
where $\boldsymbol{\ell}$ is a p-vector of positive values and Γ is a $p \times k$ matrix.

The 2nd (and most important) case can also be written as

$$k(\boldsymbol{d}) = \exp\left(-\frac{1}{2}\sum_{j=1}^{p}\frac{d_j^2}{l_j^2}\right)$$

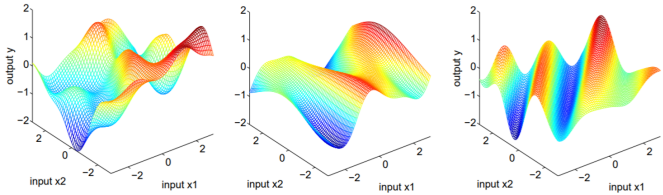
CHARACTERISTIC LENGTH-SCALE III

What is the benefit of having an individual hyperparameter ℓ_i for each dimension?



- The $\ell_1, \ldots, \ell_p$ hyperparameters play the role of **characteristic length-scales**.
- Loosely speaking, ℓ_i describes how far you need to move along axis i in input space for the function values to be uncorrelated.
- Such a covariance function implements **automatic relevance determination** (ARD), since the inverse of the length-scale ℓ_i determines the relevancy of input feature i to the regression.
- If ℓ_i is very large, the covariance will become almost independent of that input, effectively removing it from inference.
- If the features are on different scales, the data can be automatically **rescaled** by estimating $\ell_1, \ldots, \ell_p$

CHARACTERISTIC LENGTH-SCALE IV



For the first plot, we have chosen $\boldsymbol{M} = \boldsymbol{I}$: the function varies the same in all directions. The second plot is for $\boldsymbol{M} = \mathrm{diag}(\ell)^{-2}$ and $\ell = (1, 3)$: The function varies less rapidly as a function of x_2 than x_1 as the length-scale for x_1 is less. In the third plot $\boldsymbol{M} = \Gamma\Gamma^T + \mathrm{diag}(\ell)^{-2}$ for $\Gamma = (1, -1)^\top$ and $\ell = (6, 6)^\top$. Here Γ gives the direction of the most rapid variation. (Image from Rasmussen & Williams, 2006)